Thermal Physics (PH2104) Endsem Exam (December 9, 2025)

Answer ALL questions! Marks are indicated in brackets. Total marks =50

1. Using the Maxwell-Boltzmann velocity distribution $f(u)du = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2} u^2 e^{-\frac{mu^2}{2kT}} du$;

calculate (i) the mean velocity of the distribution (ii) The root mean square (RMS) velocity of the distribution. (iii) Convert the factor f(u)du in terms of energy of a particle to obtain the energy distribution f(E)dE and use it to calculate the average energy <E> of a particle. (3+3+2+2)

2. (i) Show that a biased 6-sided dice that produces outcomes 1,2,3,4,5,6 with probabilities 1/4, 1/8, 1/8, 1/8, 1/8, 1/4, has a lower entropy than a fair dice that produces outcomes with equal probabilities 1/6 for each of the 6 sides by calculating the entropy in each case. (3)

(ii) A system of particles can be distributed across 7 energy levels labelled as $E_i (i=1,..,7)$. Suppose $E_1=0$. If the temperature (T) of the system is such that $k_BT \sim 3 \times 10^{-2}$eV, and the energy associated with the different energy levels are $E_1=0$, $E_2= 3 \times 10^{-4}$eV, $E_3= 3 \times 10^{-3}$eV, $E_4= 3 \times 10^{-2}$eV, $E_5= 3 \times 10^{-1}$ eV, $E_6= 3$ eV, $E_7= 30$ eV; how many energy levels will be accessible and which is the highest accessible energy level. Justify your answer quantitatively! (3)

(iii) Two identical bodies with constant heat capacity $C_p$ at temperatures $T_1$ and $T_2$ are used as reservoirs. If both bodies remain at constant pressure, prove that the amount of work obtainable is $W= C_p (T_1+ T_2 -2 T_f)$. If the most efficient engine is used, prove that $T_f^2=T_1T_2$. (2+2)

3. (i) If $p_i$ is the probability of finding a system in the energy level $E_i$, prove that $\ln p_i = \beta(A - E_i)$ (2)

(ii) A simple harmonic oscillator has energy levels given by $E_n = (n+\frac{1}{2})\hbar\omega$ ;n=0,1, 2,...; n=0 corresponds to the ground state energy and so on. Obtain an expression for the partition function (Z) of a single simple harmonic oscillator. Prove that in the high temperature limit, $k_BT >> \hbar\omega$, the partition function (Z) is approximately given by $Z \approx (\beta\hbar\omega)^{-1}$. Obtain expressions for internal energy (U), Helmholtz free energy (A) and entropy (S), in the high temperature limit. (2+2+2+2+2)

4. Derive the third TdS equation: $TdS = C_V\left(\frac{\partial T}{\partial p}\right)_V dp + C_p\left(\frac{\partial T}{\partial V}\right)_p dV$ (3)

and show that the three TdS equations may be written as follows (2+2+2)

(a) $TdS = C_V dT + \frac{\beta T}{\kappa} dV$

(b) $TdS = C_p dT - V\beta T dp$

(c) $TdS = C_V \frac{\kappa}{\beta} dp + \frac{C_p}{V\beta} dV$

Where $\kappa$ is the *isothermal compressibility* and $\beta$ is the *volume expansivity*.

**5.** If the partition function for a single particle is given by $Z_1 = V\frac{(2\pi m k_B T)^{3/2}}{h^3}$, (i) write down the partition function ($Z_N$) for N *distinguishable* particles. (1)

(ii) Derive expressions for internal energy (U), Helmholtz free energy (A) and entropy (S) for the system of N distinguishable particles. If the system of distinguishable particles is put into 2 compartments of equal volume V such that there are N particles in each compartment, calculate the sum of the entropy in each compartment. Write down the total entropy of the system if the partition between the compartments is removed and show that the entropy is not an extensive quantity. (2+2+2+2)

<u>Formulas you may need to use</u>

Maxwell's Relations:

(i) $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$ (ii) $\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$ (iii) $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$ (iv) $\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1;$$

$$\int_0^\infty x^n e^{-Bx^2} dx = \frac{1}{2} B^{-\frac{(n+1)}{2}} \Gamma\left(\frac{n+1}{2}\right)$$

$$\int_0^\infty x^{n-1} e^{-x} dx = \Gamma(n)$$

$$\Gamma(n+1) = n\Gamma(n);\ \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

1'st TdS equation: $TdS = C_V dT + T\left(\frac{\partial p}{\partial T}\right)_V dV$

2'nd TdS equation: $TdS = C_p dT - T\left(\frac{\partial V}{\partial T}\right)_p dp$

3'rd TdS equation: $TdS = C_V\left(\frac{\partial T}{\partial p}\right)_V dp + C_p\left(\frac{\partial T}{\partial V}\right)_p dV$